

What is SoftMax Regression?

SoftMax Regression, also known as **Multinomial Logistic Regression**, is a **generalization of logistic regression** for **multi-class classification** problems.

If logistic regression is used for **binary classification**, SoftMax regression is used when your target variable has **more than two classes** (e.g., classifying an image as either cat, dog, or horse).

Key Concepts

1. Purpose

- Predict the **probability** that a given input belongs to **each possible class**.
- Output: a **probability distribution** over the classes.

2. Softmax Function

Given a vector of scores (logits) $z = [z_1, z_2, \dots, z_k]$, the softmax function computes:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

- Ensures all output values are between 0 and 1
- The sum of all softmax outputs is 1 → interpretable as **probabilities**

3. Model Output

For an input x , and weights W , biases b :

$$z = Wx + b \quad (\text{logits})$$

$$\hat{y} = \text{softmax}(z)$$

Loss Function: Cross-Entropy

Softmax is usually used with **categorical cross-entropy** loss:

$$L = - \sum_{i=1}^k y_i \log(\hat{y}_i)$$

- y_i is the **true label** (one-hot encoded)
- $\hat{y}_i$ is the **predicted probability**

This loss encourages the model to assign **high probability to the correct class**.

Example

Imagine you're classifying handwritten digits (0–9). So, you have **10 classes**.

- The model outputs 10 scores (logits), one per class.
- SoftMax converts these scores into probabilities (like [0.05, 0.01, 0.8, ..., 0.01]).
- The predicted class is the one with the **highest probability**.

Sample Prediction

Suppose the model gets an image of a handwritten "3". The raw output (logits) might be:

logits = [2.3, 1.1, -0.5, 5.6, 0.2, -1.1, 0.3, -0.2, 1.7, -0.8]

SoftMax turns this into:

probabilities = [0.02, 0.01, 0.005, 0.9, 0.01, 0.002, 0.01, 0.003, 0.02, 0.002]

Predicted class = 3 (highest probability = 0.9)